Patch-Level Selective Semi-Amortized Inference Experiment Summary

```
Generated: 2026-03-19 13:23

Study: selective_semiamortized_patch_fullstudy_leaves_18p7_v2
Root: /cluster/VAST/kazict-lab/e/lesion_phes/code/segmentors_backbones/experiments/exp_v1/outputs/selective_semiamortized_patch_fullstudy_leaves_18p7_v2

Key parameters:
  Patch size: 224x224, stride: 112
  Lesion-aware selection: nonzero_fraction >= 0.10
  Oracle: 16 random starts + 8 refinement steps
  Features: DINOv2 (384d), ResNet-18 (512d), Handcrafted (45d)
  Models: Ridge, MLP(128,64), 5-member ensemble
  Refinement budgets: direct/tiny(3)/small(6)/medium(10)/large(20)/full(24)
  Aggregation: 6 methods (mean, unc-weighted, lesion-weighted, top-k, selective, median)
  Splits: train=104, val=26, test=44, OOD=14, external=63
```

Pipeline Status

```
Pipeline Stage Completion Status:

  setup                      completed   2026-03-18T18:52:55  directories created, v1 manifests reused
  patch_selection            completed   2026-03-18T18:56:51  1059 patches from 174+63 leaves
  splits                     completed   2026-03-18T18:58:45  train=104, val=26, test=44, ext=63
  oracle                     completed   2026-03-18T19:49:14  leaves=870, ext=187
  features                   completed   2026-03-18T20:33:40  n=1057, dino=True, resnet=True
  ablations                  completed   2026-03-18T20:35:11
  models                     completed   2026-03-18T20:50:29  primary=dino, backbones=['handcrafted', 'dino', 'resnet']
  refinement                 completed   2026-03-18T22:01:13  2442 results across 6 budgets
  selective_routing          completed   2026-03-19T06:31:32  136 policies, best speedup=11.1x
  aggregation                completed   2026-03-19T06:31:41  44 leaves, 264 rows
  evaluation                 completed   2026-03-19T06:33:26
  geometry                   completed   2026-03-19T06:33:39
  figures_tables             completed   2026-03-19T06:33:46
```

Patch Dataset Overview

```
Total selected patches: 1059
From leaves domain: 870
From 18.7 domain: 189
Unique parent leaves (leaves): 174
Unique parent leaves (18.7): 63
```

Oracle Optimization Results

```
Total oracle runs: 1057
Successful: 1057
Failed: 0
Mean score: -946064.8127
Mean runtime: 2.5s
Mean evaluations: 22

Per-parameter oracle statistics (mean +/- std):
  mu                      0.2332 +/- 0.1610
  lambda                  0.3345 +/- 0.1631
  diffusion_rate          0.1783 +/- 0.0935
  alpha                   0.0261 +/- 0.0148
  beta                    0.1288 +/- 0.0721
  gamma                   0.2076 +/- 0.1240
  energy_threshold        40.8997 +/- 14.1707
```

Full Evaluation Summary

split	method	parameter	mae	rmse	r2
test_leaves	direct_ensemble	mu	0.2594	0.3159	-2.6398
test_leaves	direct_ensemble	lambda	0.4352	0.5514	-10.8132
test_leaves	direct_ensemble	diffusion_rate	0.2389	0.3089	-9.9132
test_leaves	direct_ensemble	alpha	0.4335	0.4987	-1198.9
test_leaves	direct_ensemble	beta	0.2150	0.2791	-15.6348
test_leaves	direct_ensemble	gamma	0.6195	0.7147	-37.5311
test_leaves	direct_ensemble	energy_threshold	39.5983	42.0638	-7.7991
test_leaves	direct_ensemble	AGGREGATE	5.9714	6.3904	-183.3153
test_leaves	direct_ridge	mu	0.2350	0.2985	-2.2485
test_leaves	direct_ridge	lambda	0.2465	0.3224	-3.0385
test_leaves	direct_ridge	diffusion_rate	0.1530	0.2030	-3.7112
test_leaves	direct_ridge	alpha	0.0228	0.0294	-3.1581
test_leaves	direct_ridge	beta	0.1004	0.1331	-2.7843
test_leaves	direct_ridge	gamma	0.1734	0.2264	-2.8653
test_leaves	direct_ridge	energy_threshold	20.5155	26.7585	-2.5608
test_leaves	direct_ridge	AGGREGATE	3.0638	3.9959	-2.9095
test_leaves	direct_mlp	mu	0.7027	0.8778	-27.0974
test_leaves	direct_mlp	lambda	0.4395	0.5783	-11.9916
test_leaves	direct_mlp	diffusion_rate	0.9919	1.1331	-145.8074
test_leaves	direct_mlp	alpha	1.2639	1.6091	-12492.9
test_leaves	direct_mlp	beta	0.5366	0.7025	-104.3946
test_leaves	direct_mlp	gamma	0.5296	0.7189	-37.9768
test_leaves	direct_mlp	energy_threshold	40.2740	42.6642	-8.0520
test_leaves	direct_mlp	AGGREGATE	6.3912	6.8977	-1832.6
test_leaves	constant_mean	mu	0.1418	0.1656	0.0000
test_leaves	constant_mean	lambda	0.1385	0.1604	0.0000
test_leaves	constant_mean	diffusion_rate	0.0817	0.0935	0.0000
test_leaves	constant_mean	alpha	0.0122	0.0144	0.0000
test_leaves	constant_mean	beta	0.0584	0.0684	0.0000
test_leaves	constant_mean	gamma	0.0908	0.1151	0.0000

Patch vs Full-Leaf Aggregation

method	mean_leaf_mae
lesion_weighted	10.0216
mean_direct	10.0211
robust_median	10.0271
selective_mixture	7.9892
topk_confident	10.0355
uncertainty_weighted	10.0225
fullleaf_selective_baseline	0.0715

Within-Leaf Spatial Heterogeneity

```
Leaves with multi-patch oracle: 174
Mean within-leaf param std: 1.7020
Mean within-leaf param range: 4.5926
Max within-leaf param std: 23.2671

Distribution of within-leaf heterogeneity:
  Q25: 1.1629
  Q50: 1.6344
  Q75: 2.2525
  Q90: 2.5743
```

Fig01 Runtime Quality Pareto

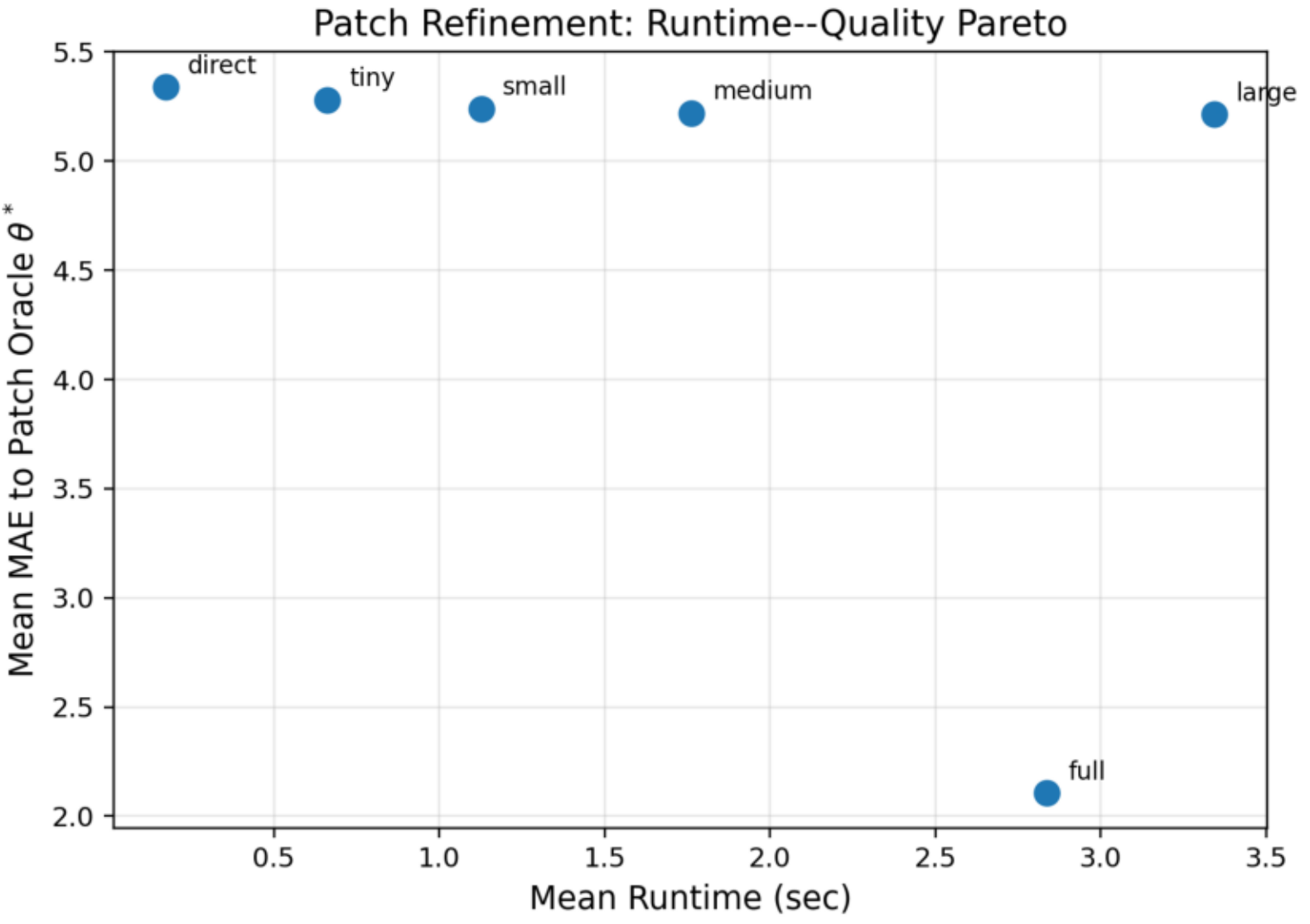


Fig02 Aggregation Comparison

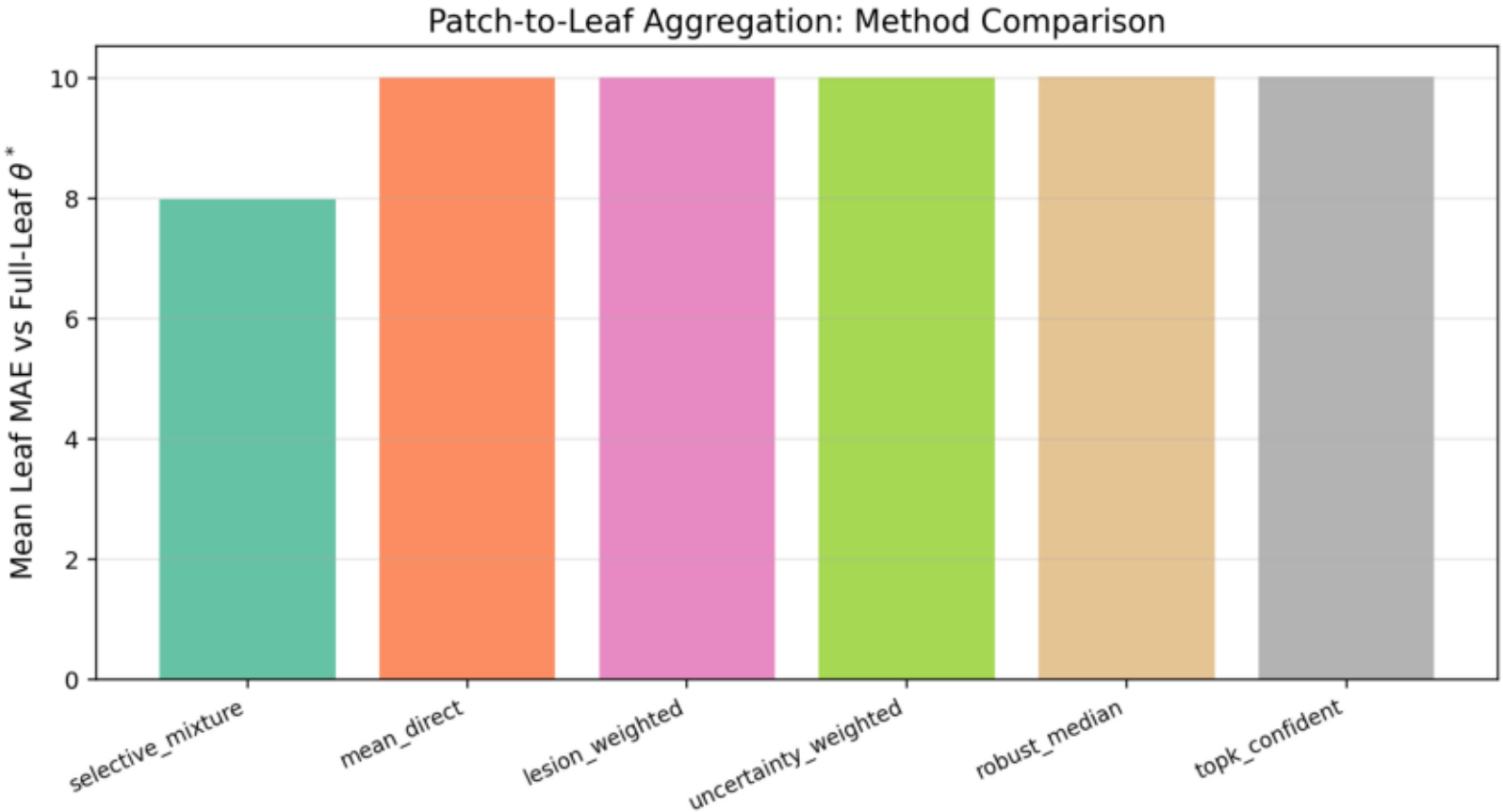


Fig03 Risk Coverage

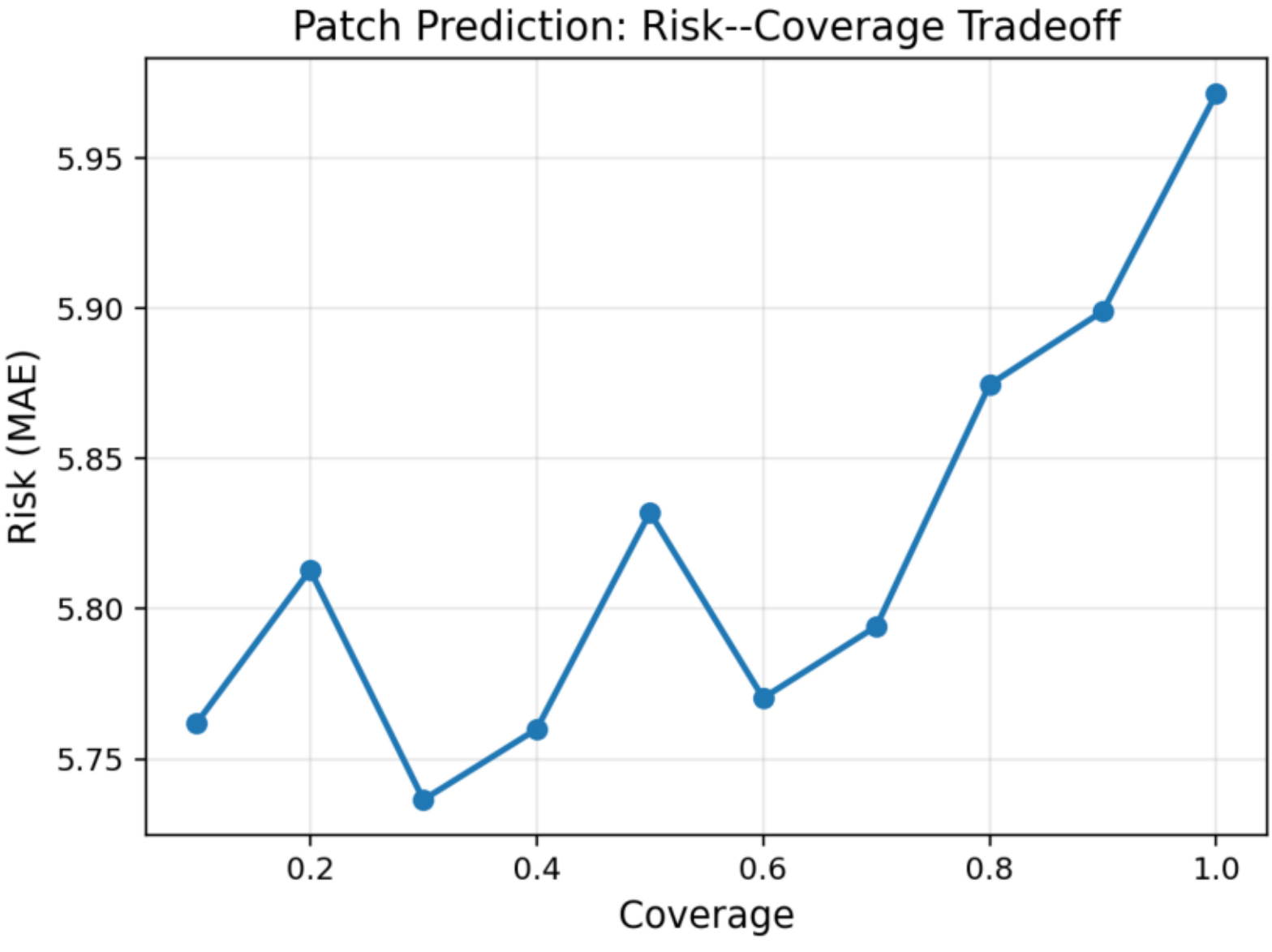


Fig04 Calibration

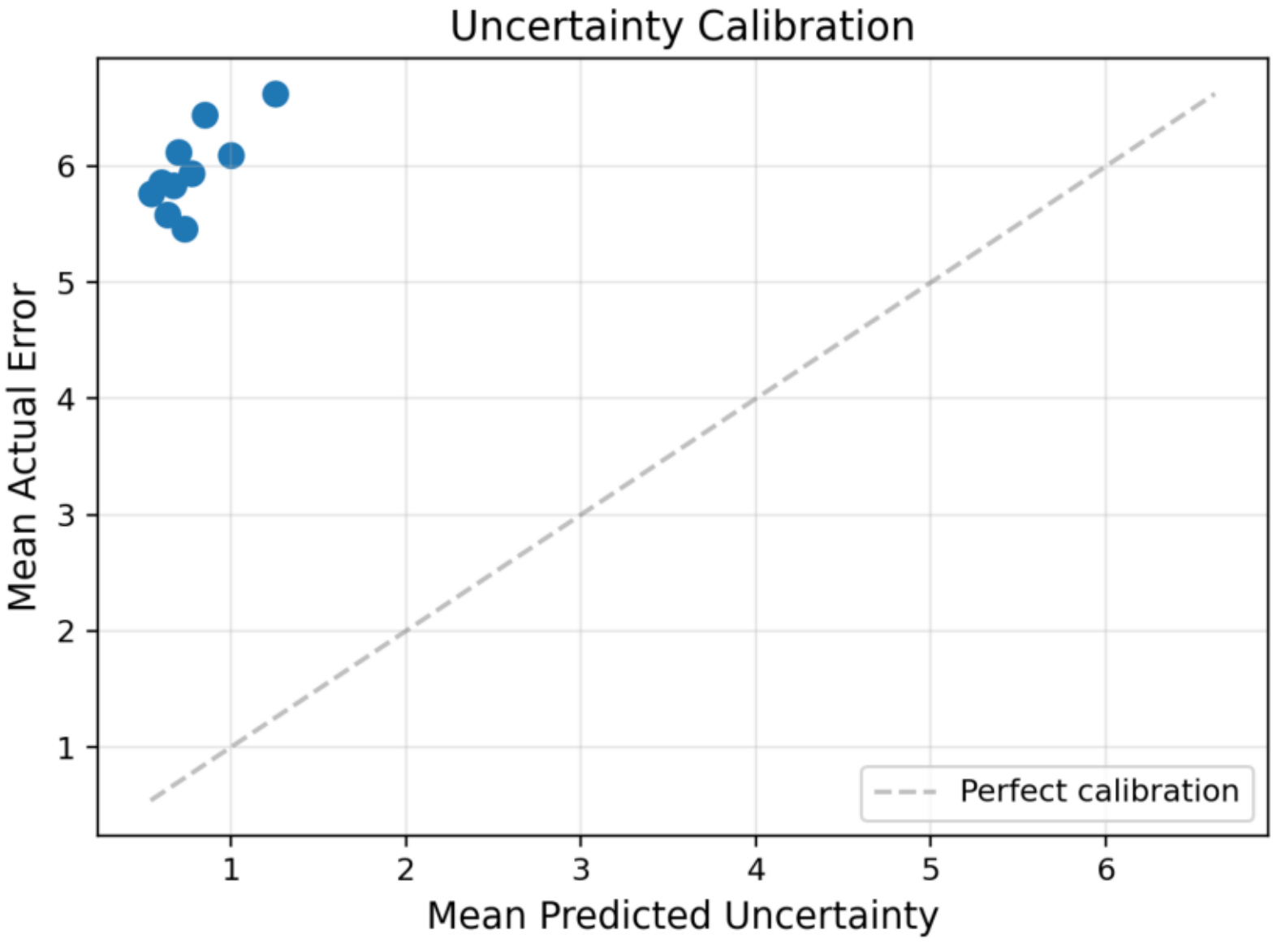


Fig05 Geometry Pca

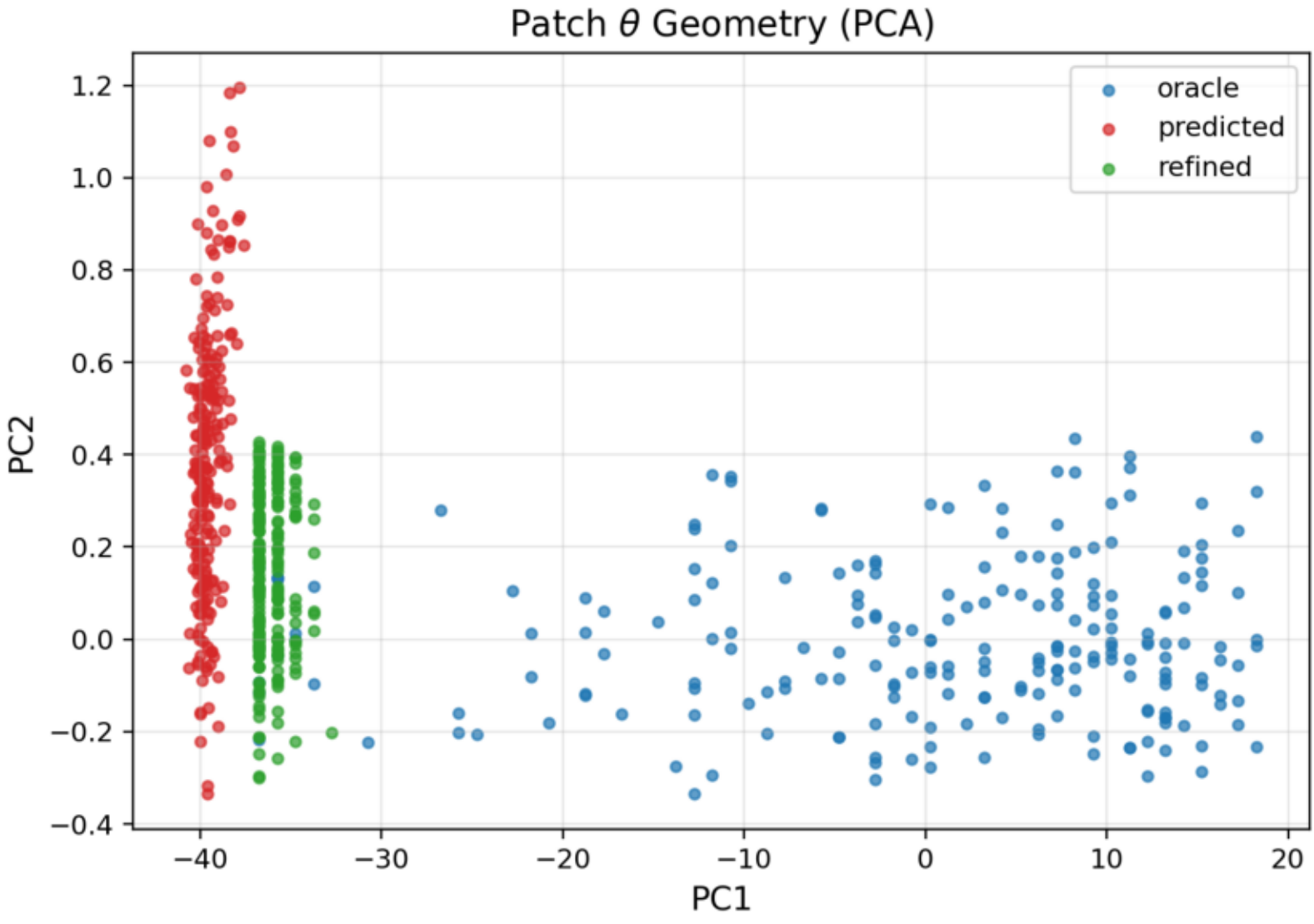


Fig06 Uncertainty Vs Error

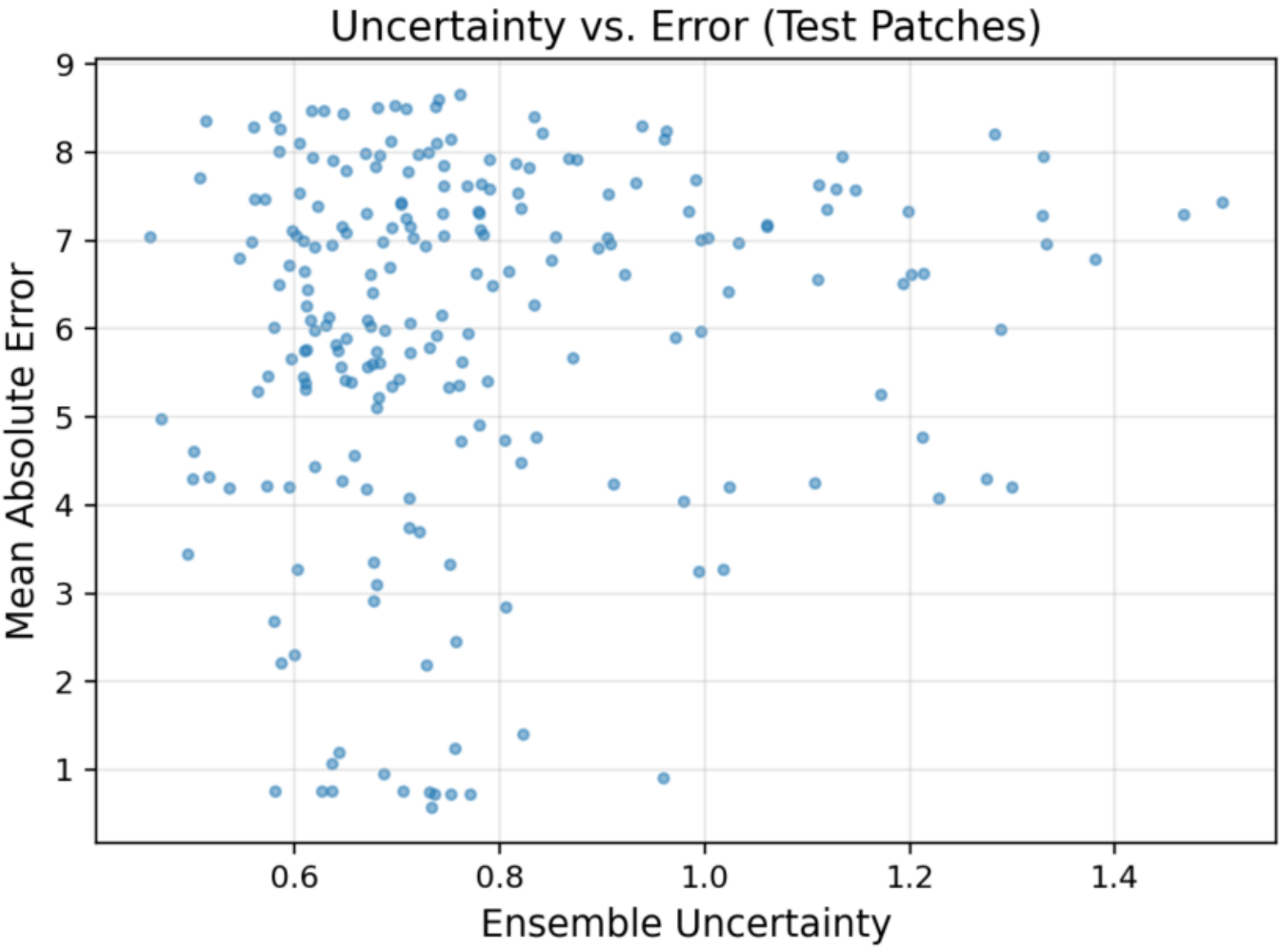


Fig07 Spatial Heterogeneity

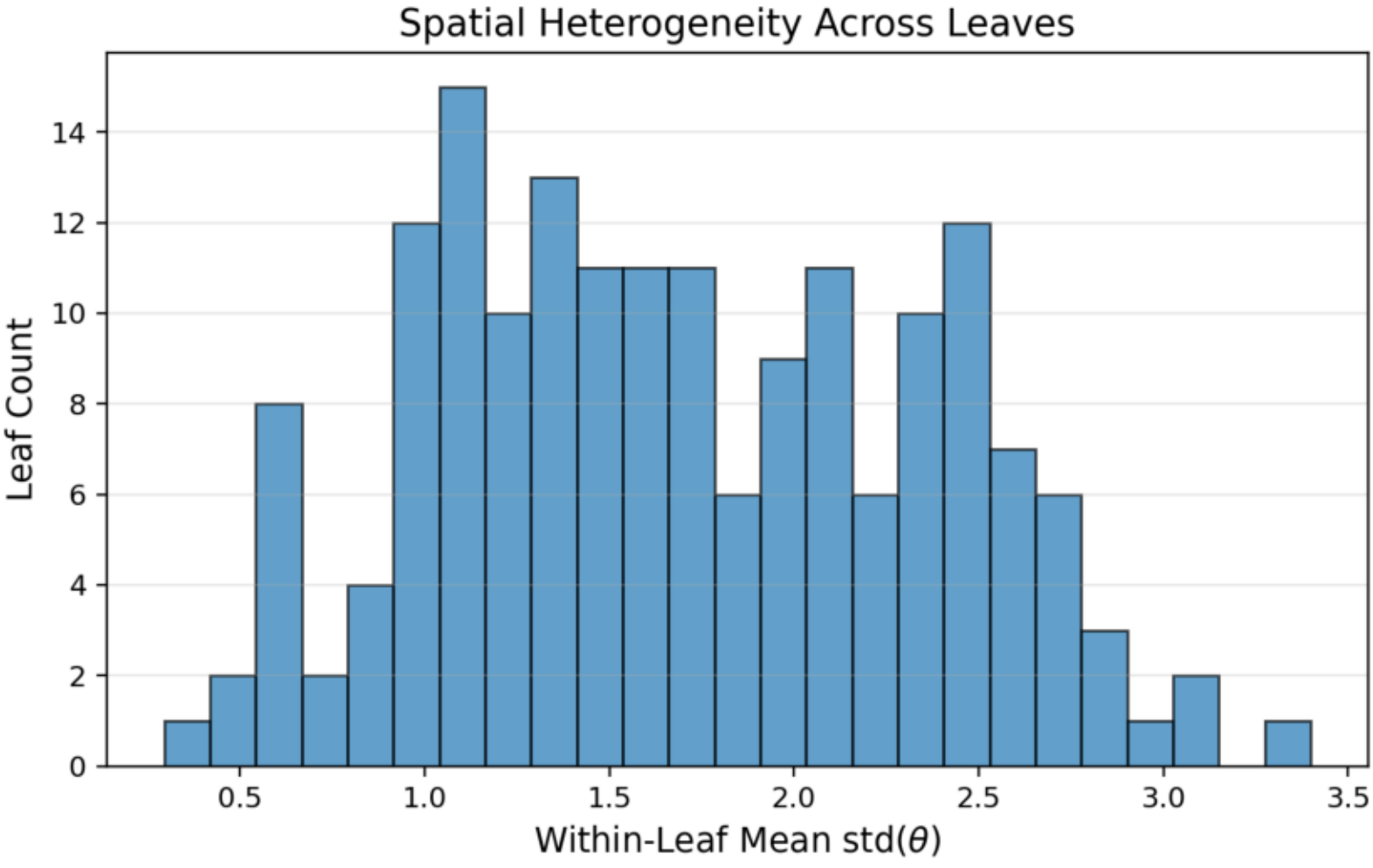


Fig08 Domain Robustness

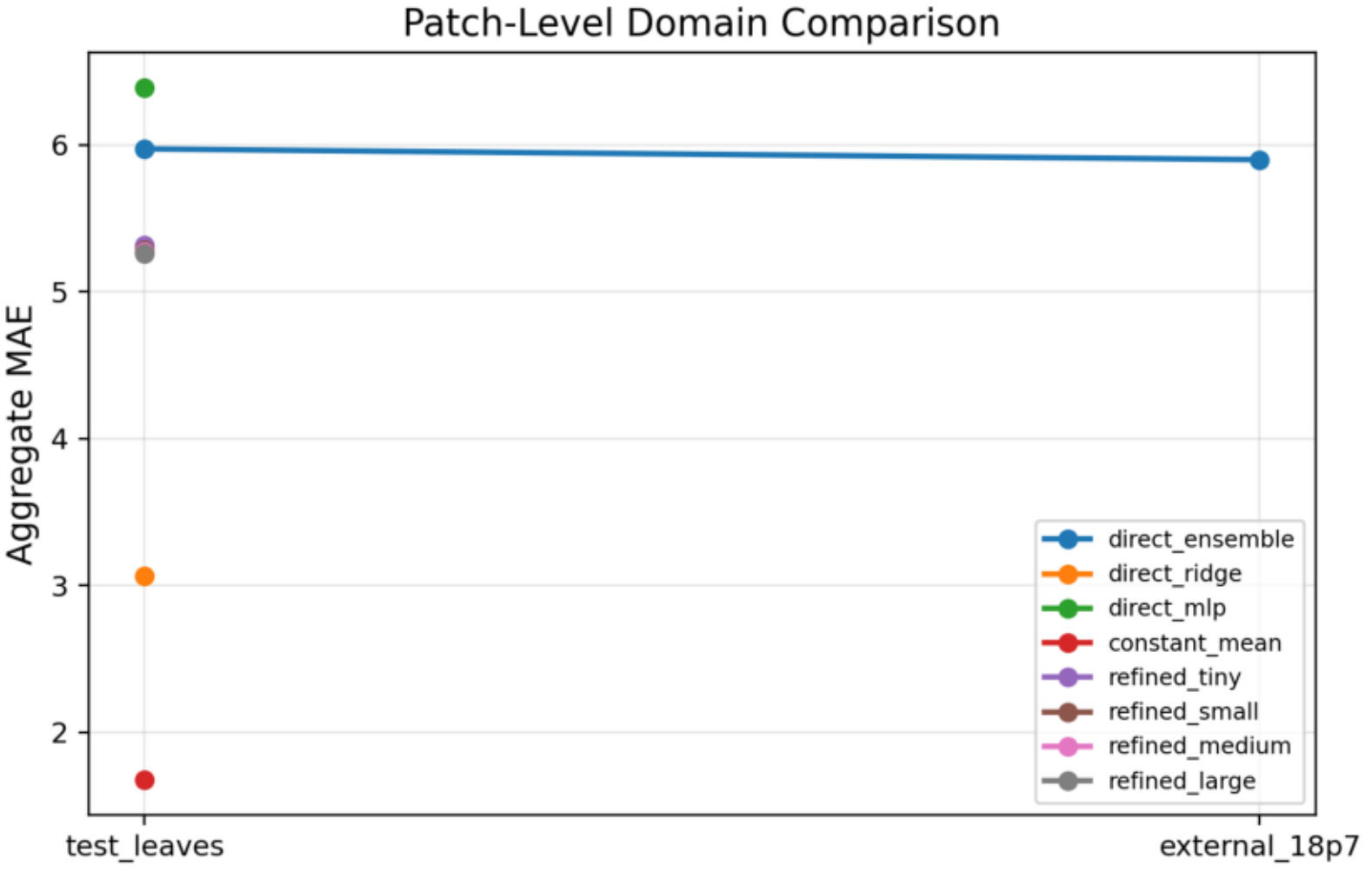


Fig09 Per Parameter Difficulty

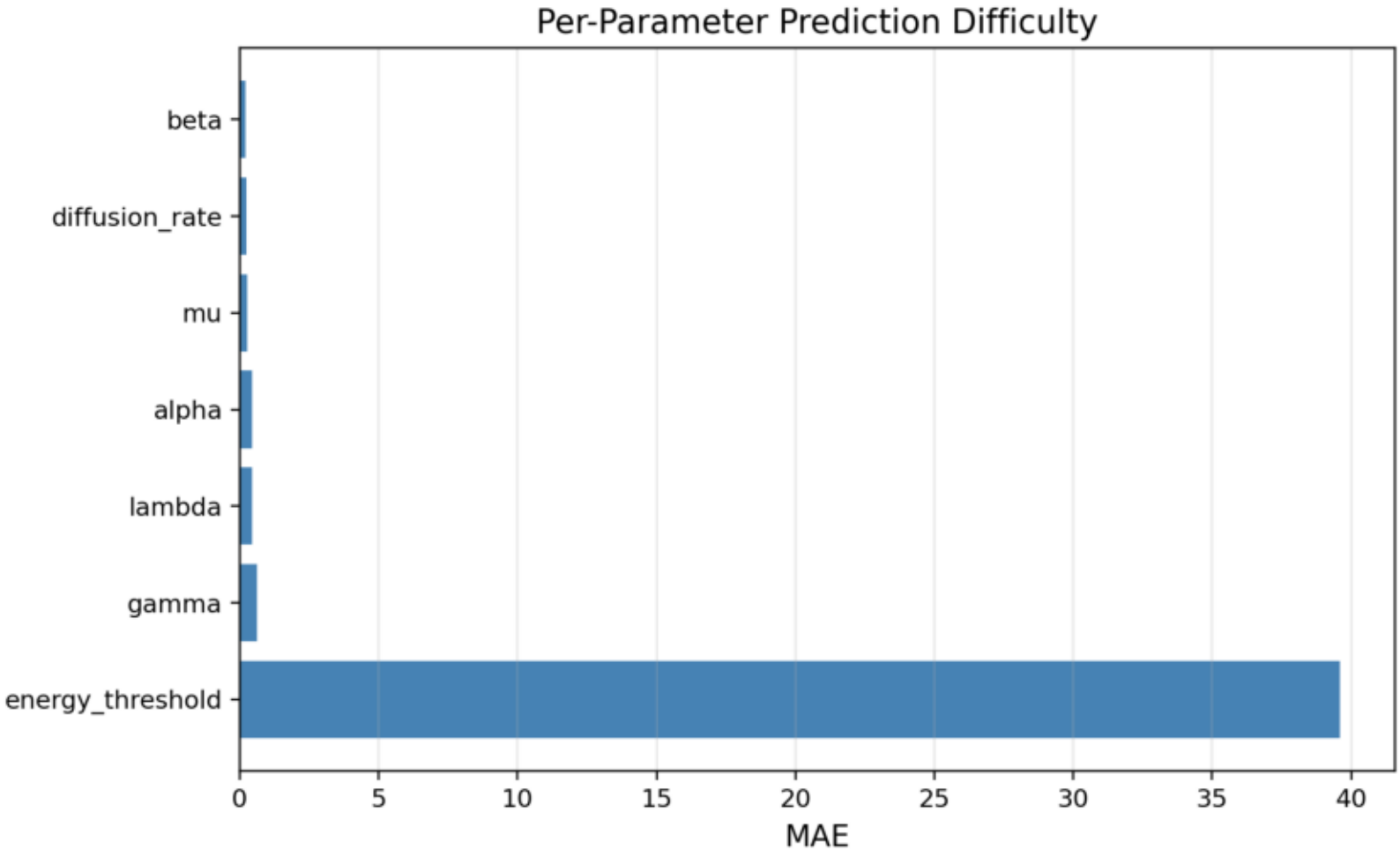


Fig10 Performance Comparison

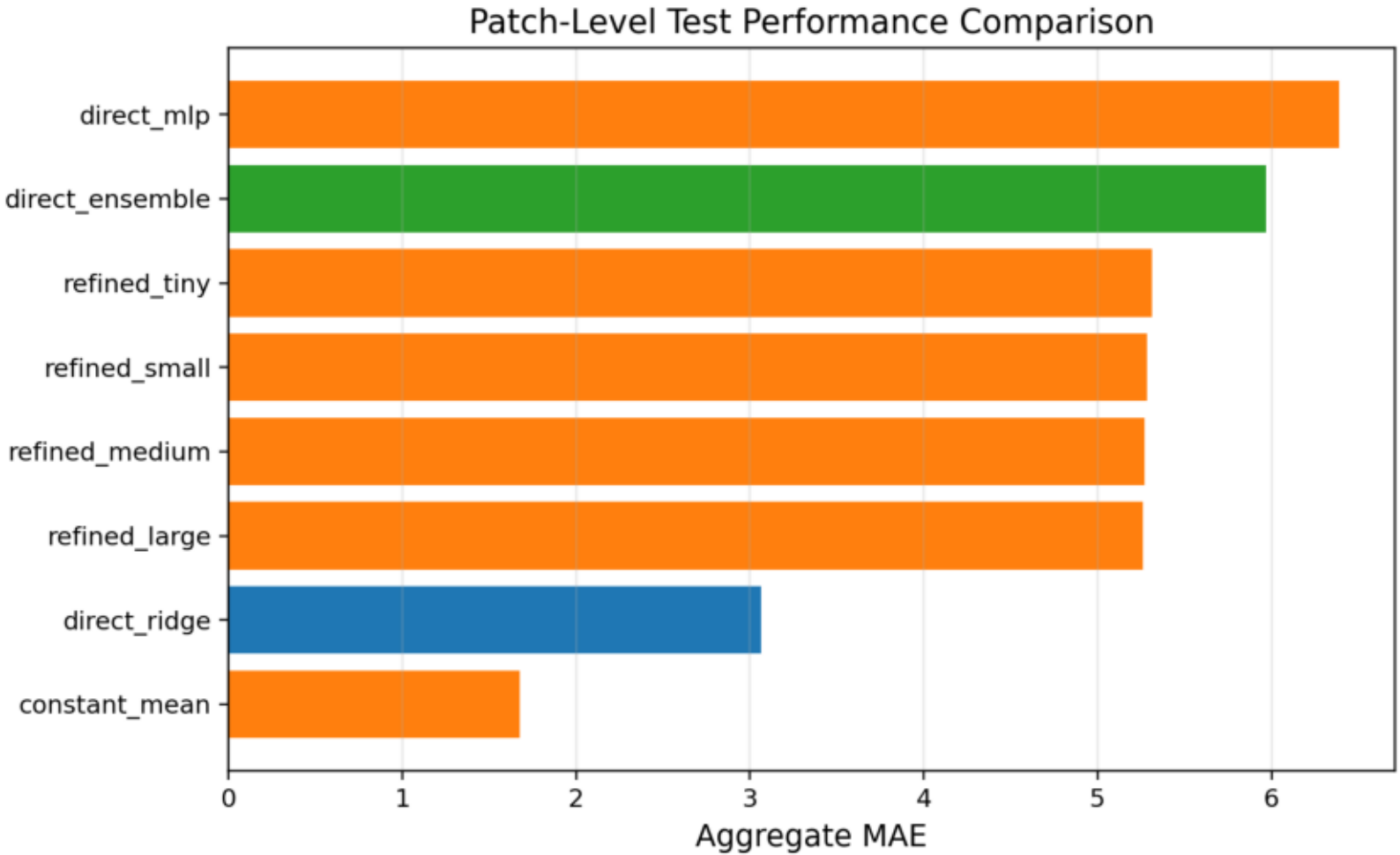
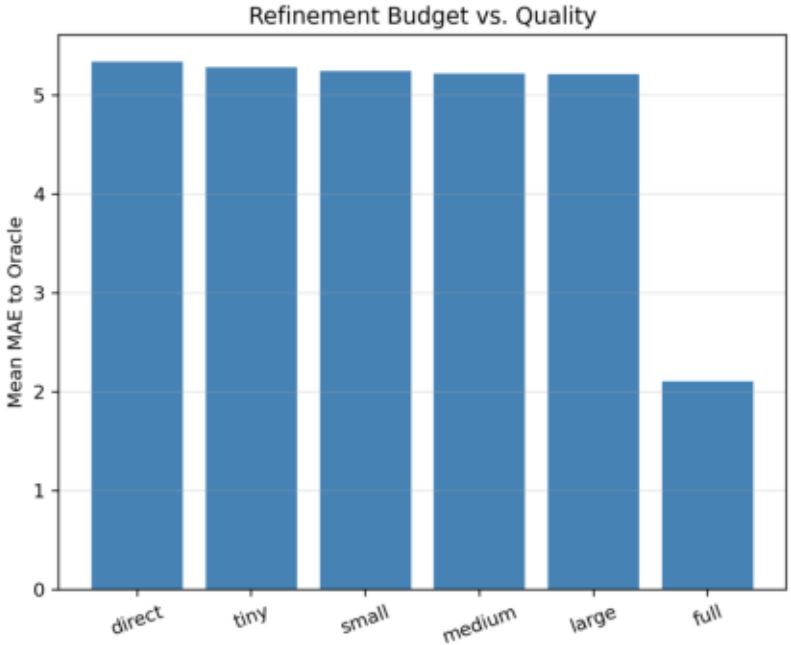


Fig11 Refinement Budget



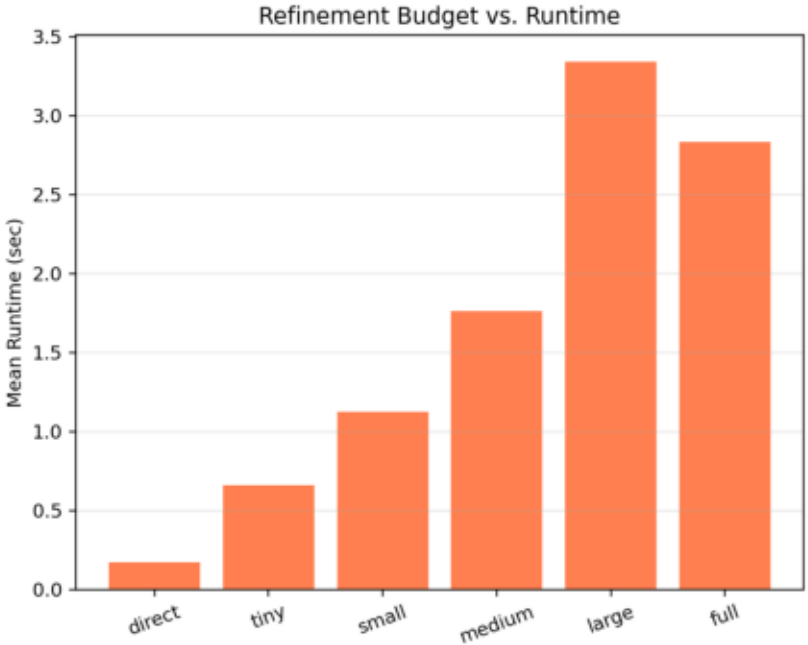
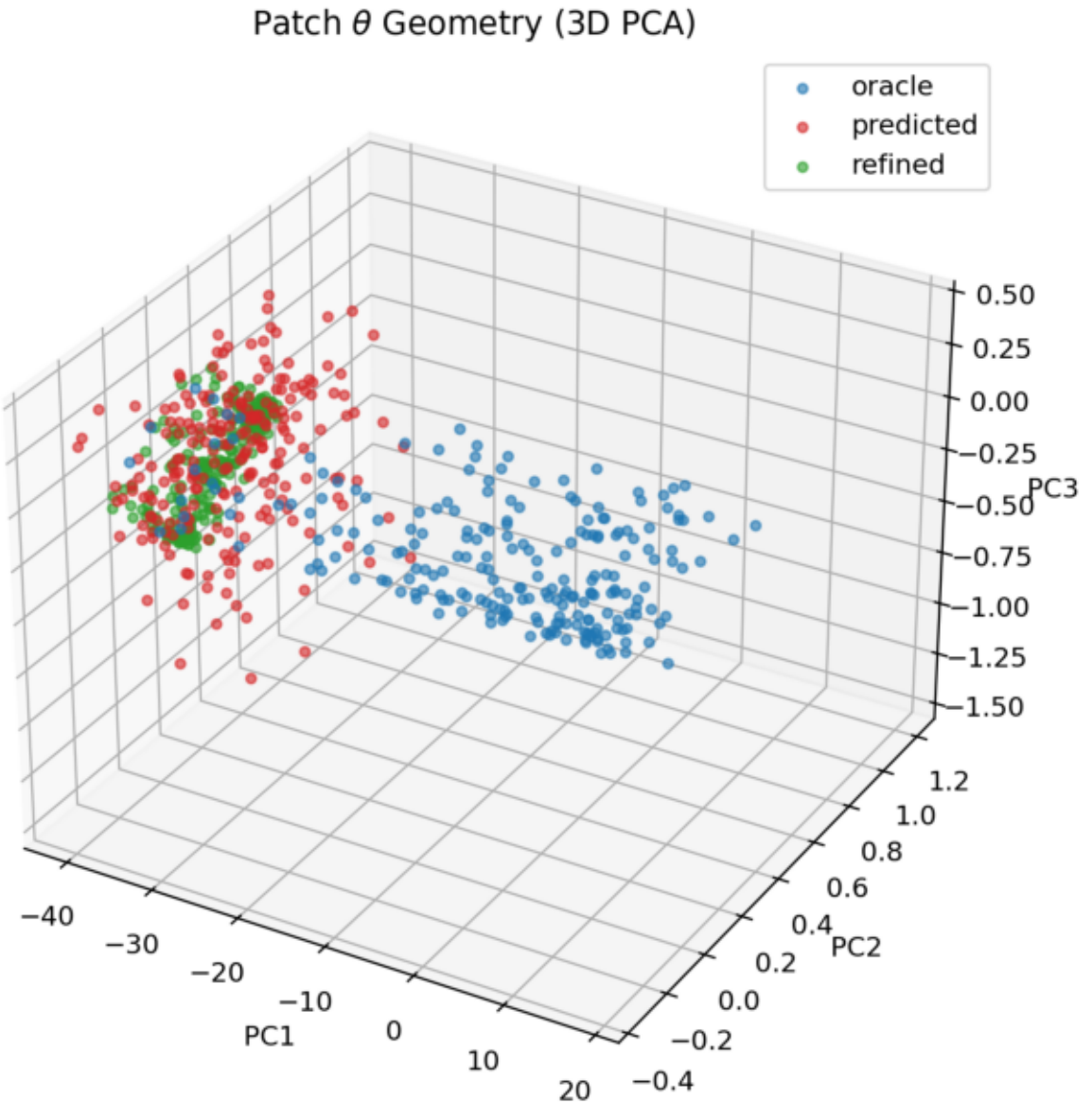


Fig12 Geometry 3D



Artifact Provenance

artifact	source	reused_or_new	notes
patch_manifest_all.csv	v1 study	reused	300286 entries from dense+lesionaware extra
patch_manifest_dense.csv	v1 study	reused	dense grid patches
patch_manifest_lesionaware.csv	v1 study	reused	152667 lesion-aware patches
domain_summary.csv	v1 study	reused	domain statistics
patch_quality_report.txt	v1 study	reused	quality filtering statistics
leaf_inventory.csv	v1 study	reused	174 leaves + 63 external
selected_patch_set.csv	v2 pipeline	new	1059 patches (870 leaves + 189 ext)
leaf_level_split.csv	v2 pipeline	new	train=104 val=26 test=44
patch_level_split.csv	v2 pipeline	new	inherits from leaf split
ood_split.csv	v2 pipeline	new	14 OOD leaves from test
domain_split.csv	v2 pipeline	new	leaves vs 18.7
learning_curve_subsets.csv	v2 pipeline	new	10-100% training fractions
external_only_split.csv	v2 pipeline	new	63 external leaves
patch_opt_summary_all.csv	v2 pipeline	new	real oracle optimization with 16+8 budget
handcrafted_patch_features.npy	v2 pipeline	new	color/texture/gradient features
dino_patch_features.npy	v2 pipeline	new	DINOv2-ViT-S/14 frozen features (if torch availa
resnet_patch_features.npy	v2 pipeline	new	ResNet-18 frozen features (if torch available
all_models.pkl	v2 pipeline	new	Ridge+MLP+Ensemble per feature set
patch_predictions_leaves.csv	v2 pipeline	new	predictions on primary domain
patch_predictions_18p7.csv	v2 pipeline	new	predictions on external domain
patch_refinement_casewise_results.csv	v2 pipeline	new	real scorer refinement at 5+ budgets
patch_selective_policy_results.csv	v2 pipeline	new	threshold sweep results
leaf_aggregation_results.csv	v2 pipeline	new	6 aggregation methods comparison
full_evaluation_summary.csv	v2 pipeline	new	comprehensive evaluation metrics
references_patch_v2.bib	v2 new	new	extended bibliography with patch-specific refer
main_patch_fullstudy_v2.tex	v2 new	new	full manuscript